

CSC2062 – Assignment 1

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Introduction

All the work that have been done to complete this assignment will be described, justified and results will be interpreted.

Section 1

A total of 112 images have been drawn using the recommended software GIMP. 14 images for every required object (envelope, golfclub, pencil, wineglass, cherry, banana, lemon and tree) with 45*45 pixels resolution, with grayscale colours only and drawn by a fully black colour. When created they were named according to the object name followed by its index -as shown in PGM Folder- to minimize the time spent in creating the database and saved as PGM files. Using the python code(section1_code.ipynb) to convert the PGM images into a matrices as CSV files, convert every black pixel into 0 and whites into 1s and rename the files as required. In section1_code.ipynb Pillow Library was used to read the PGM files, GLOB to return a list of all PGM files paths, NumPy to convert images from objects into matrices and convert the grayscale into 0s and 1s, OS to extract filenames and to construct the output file paths and re to extract the object name and index.

Section 2

In section2_code.ipynb, a python code was written to extract the 16 feature as shown in task 2.1 and to save them in a CSV file (40450780_features.csv). A pandas's DataFrame with 18 named columns was used to save the extracted features, label and index for every image. Its methods was also used later to sort the data according to the alphabetical order firstly and index secondly and to save the data in a CSV file after the extraction process. As for the extraction, label and index were extracted from the file name using the split method. An image was loaded into a NumPy array for the extraction of the features. For number of black pixels(nr_pix), NumPy sum method was used to calculate the sum of all black pixels in an image represented by 0 and a similar method was used to extract the number of rows/columns with exactly 1, with exactly 2 and with 3 or more black pixels(rows_with_1, cols_with_1, rows_with_2, cols_with_2, rows_with_3p, cols_with_3p). Using "for loop" to iterate over rows/columns and if statement to enforce the condition. For the height and the width, NumPy's "where method" was used to collect the index of the row/column for every black pixel and save it into a NumPy array then the height/width is the difference between the maximum and minimum. The number of black pixels in the row/column that has the most black pixels(maxrow and maxcol) were calculated using NumPy's "sum method" to calculate the sum of black pixels in each row/column and save it in an array. Using the "max method" it returns the intended value. Using SciPy's "ndimage.label()" to calculate the number of black connected areas(connected_areas), number of eyes(eyes) and the hollowness. For the connected areas the only parameter used is image == 0. Which will change the matrix into trues and falses (1s and 0s) and calculate the number of connected true (1s/ black) areas. It returns a matrix of the image showing the connected areas -was not used in this part- and the number of the connected areas which was used. As for the eyes, the parameters were image==1 for white

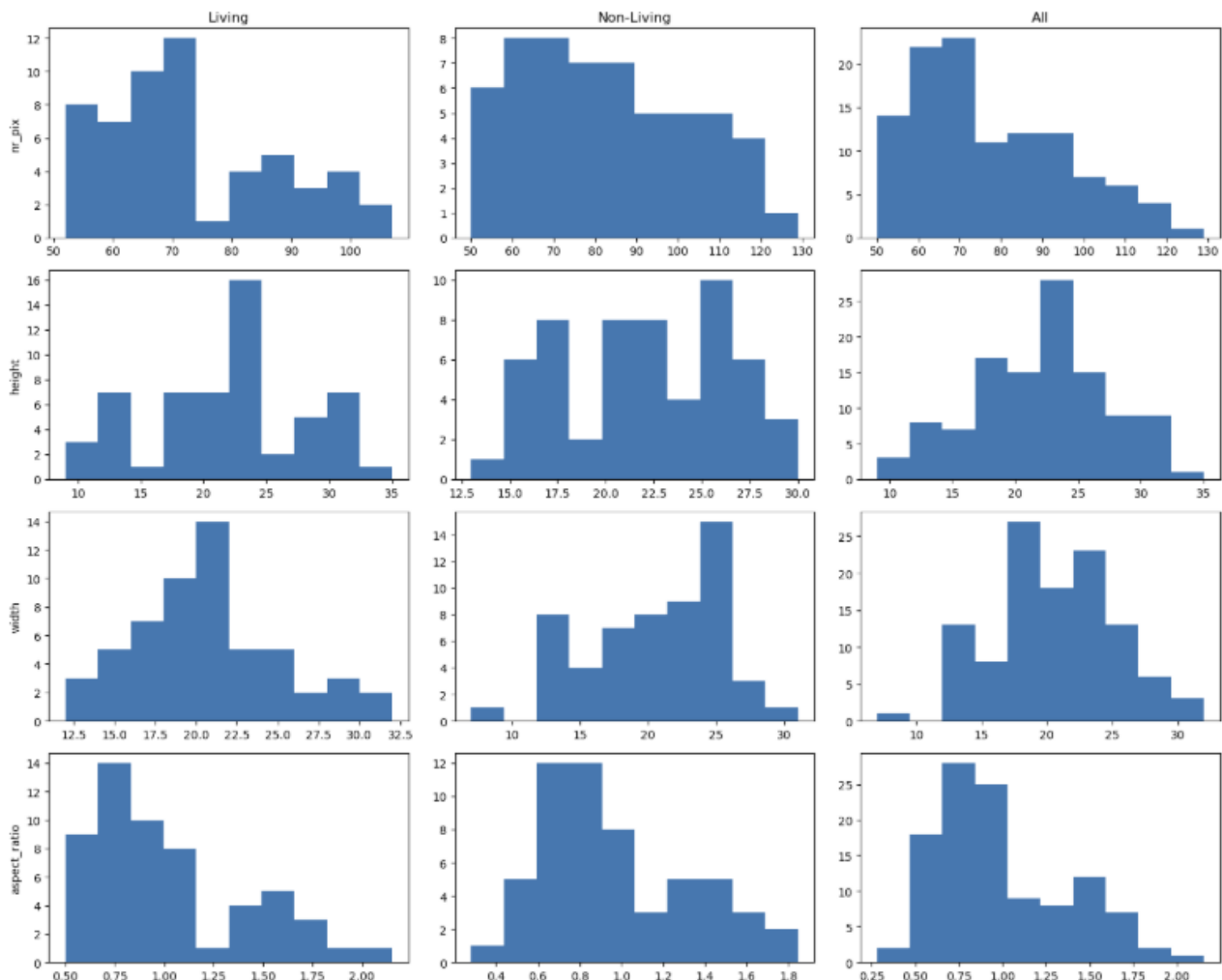
pixels and the structure matrix to define the what white connected area(eye) is, then it was subtracted by 1 for the background. For hollowness, NumPy's "sum methos" was used to calculate the number of white connected pixels -in the eyes- excluding the background using the condition labelled_white > 1 as the background will always be labelled 1 in our case. Its then divided by the number of black pixels. For the custom feature, the density of black pixels to the total area of the actual image was to chosen. It is a unique feature that could be useful to differentiate between object. Logically, image with more details will have more density.

Section 3

In this section a statistical analytics was performed as shown in section3_code file. Starting by loading the features file into a Pandas's DataFrame and two sub-data Frames to hold living objects and non-living objects separately.

Section 3.1

Using Matplotlib's "subplots method" to plot the intended histograms as subplot as showing in figure 1.1.



The histograms of the number of black pixels shows that the data for all objects are right skewed as for the living objects most images have 60-80 black pixels and skewed to the right as well. The non-living objects have a similar peak between 60 and 80 however it spreads out to up to 130.

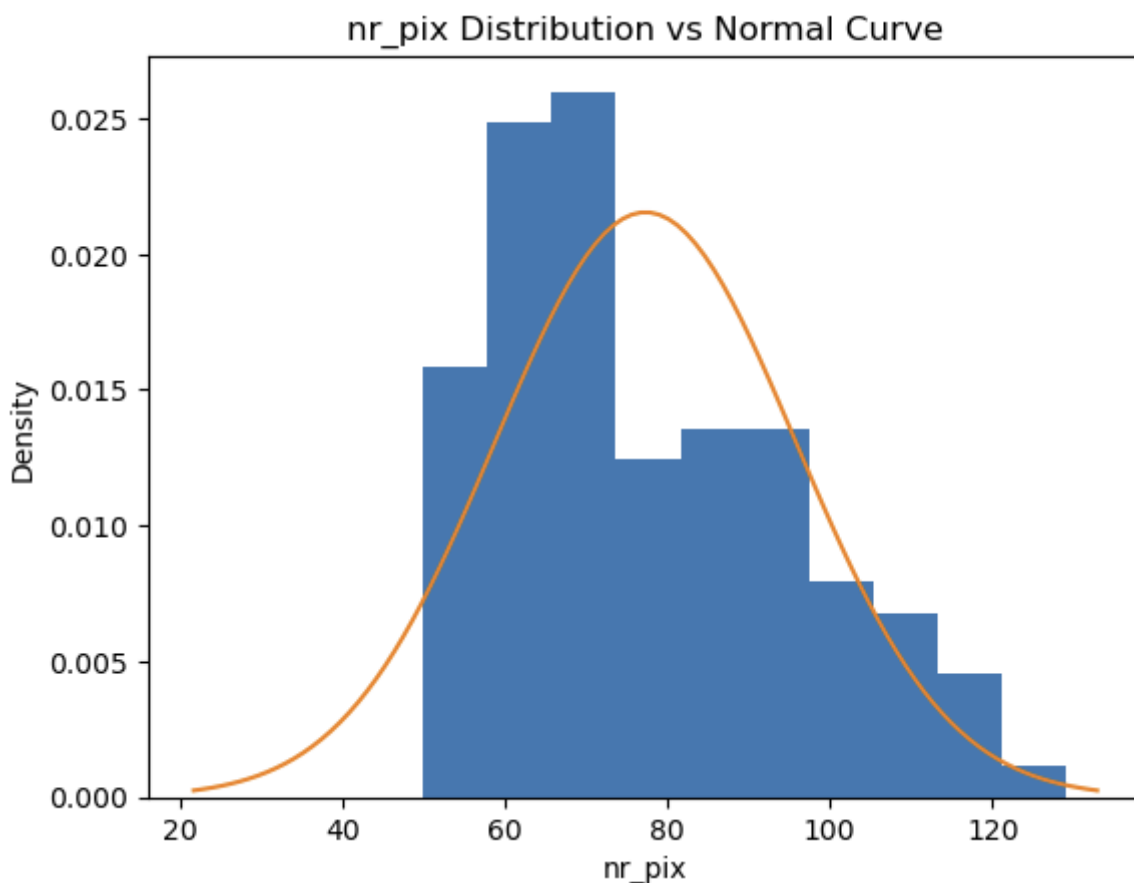
The distribution of the hight for all object is roughly normal and peaks between 20 and 25 and for living objects as well however it appears to be even rougher. For the non-livings it shows a 3 peaks.

Livings appears roughly normal and the combined as well however the living peaks at 20-22 and the combined at 17.5-2. As for the non-livings the appear to be left skewed. It could be an effective feature to be used to distinguish between them.

For the width/hight ratio they appears to be all right skewed varying in the strength.

Section 3.2

After calculating the mean and the variance which resulted in 77.31 and 342.97 respectively. The theoretical normal distribution was generated using `norm.pdf()` method using the mean and standard deviation and an array containing 100 theoretical normal samples generated by NumPy's `linspace()` method and then it was plotted correspondingly with the number of black pixels density histogram as shown in figure 1.2



The normal curve does not fit well with the data. The data peaks before the theoretical normal distribution and drops at approximately 50 while the normal extends further.

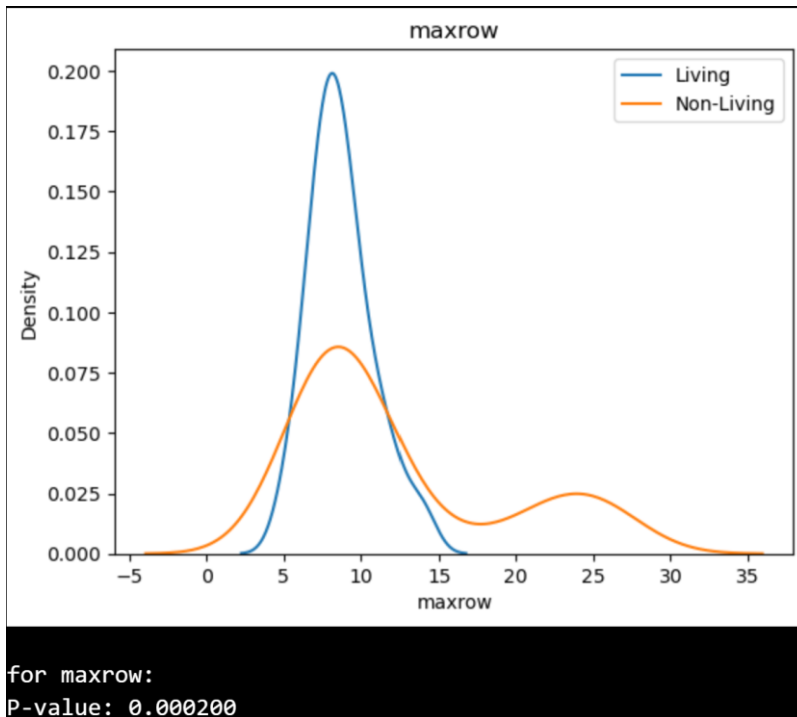
Section 3.3

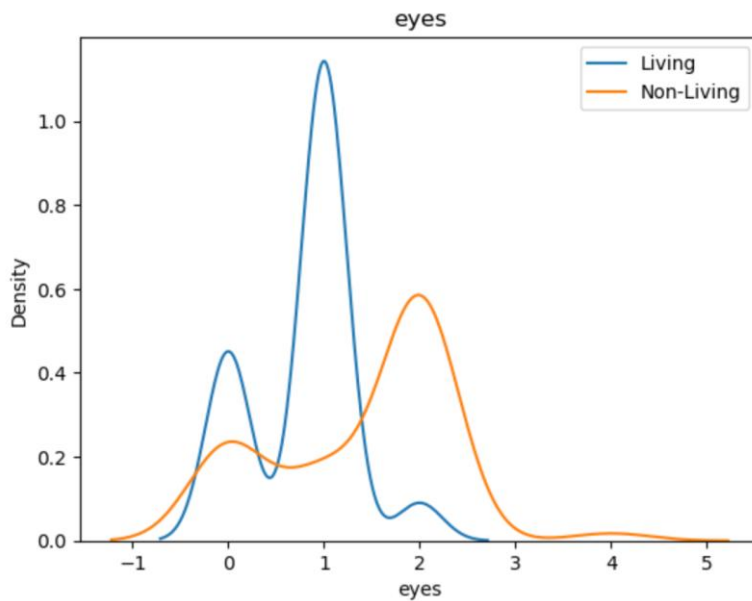
Using `norm.ppf()` to calculate the cutoff value which resulted in 107.77. This seems to be aligned with actual data.

Section 3.4

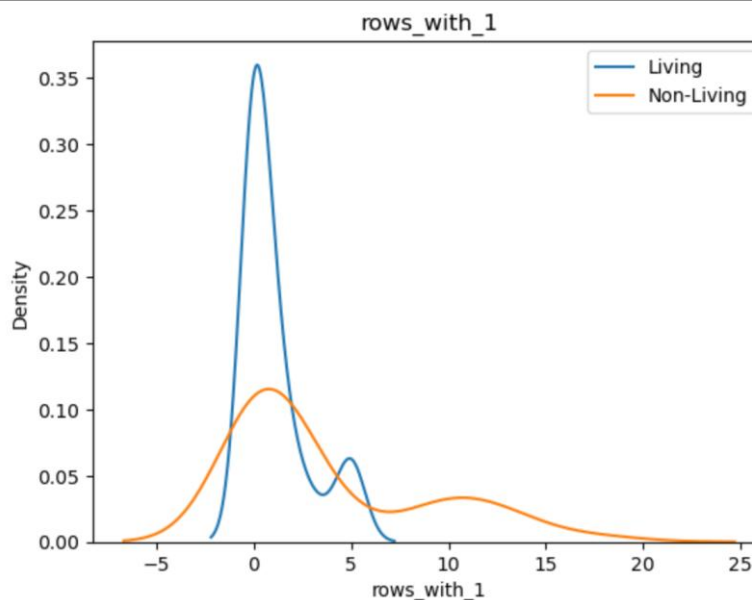
To preform such analyses and prove that some features show a significant difference between living and non-living, Null hypothesis was conducted assuming the data is approximately normally distributed. The null hypothesis(H_0) assumed no difference between the groups and it was rejected for 10 features where the p-

values was below the significant level 0.05. using “ttest_ind() method” to return the p-value for each feature and Seaborn’s “kdeplot() method” to plot visualisations which what is seen as the most useful and expressing however it is not accurate as it shows values below zero which is imposable in the sampled data. The reason is, it estimates a smooth curve based on the data.





```
for eyes:
P-value: 0.000021
```



```
for rows_with_1:
P-value: 0.000171
```

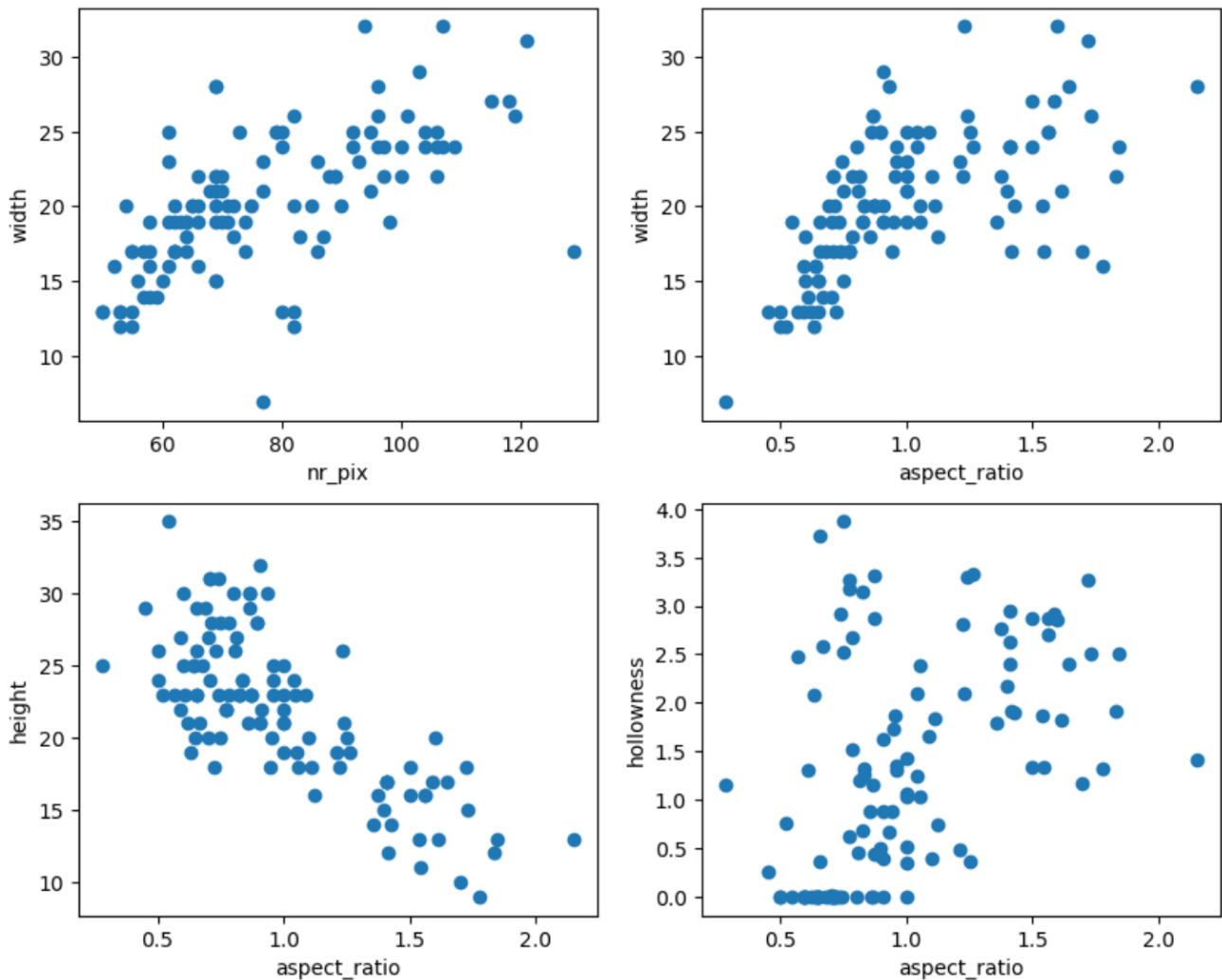
In these three figures are the features with lowest p-value. As it shown in the figures there is a clear separation between the two groups. The eyse for example, livings predominantly have 1 eye while non-livings tend to have 2 eyes.

Section 3.5

Using pandas'a "corr()" method" to calculate the Pearson correlation coefficient between the intended features and save them in matrix as shown in the figure below

	nr_pix	height	width	aspect_ratio	hollowness
nr_pix	1.000000	0.146095	0.630543	0.298516	0.219918
height	0.146095	1.000000	0.004254	-0.755253	-0.435907
width	0.630543	0.004254	1.000000	0.606663	0.279276
aspect_ratio	0.298516	-0.755253	0.606663	1.000000	0.471501
hollowness	0.219918	-0.435907	0.279276	0.471501	1.000000

And with metaplot's "subplots() methos" the pairs of features with highest correlations was plotted



The calculation resulted that the aspect ratio and hight were the most corelated with a corelation coefficient -0.755 which is strong negative correlation because aspect ratio is inversely proportionate to hight as it is equal to width/hight.

Section 4

Section 4.1

Linear regression was chosen because the correlation analysis has showed a strong linear relationship between aspect ratio and the hight and width. Using only these two feature have resulted in an $R^2 = 0.942$

Section 4.2

OLS Regression Results						
=====						
Dep. Variable:	aspect_ratio		R-squared:	0.942		
Model:	OLS		Adj. R-squared:	0.941		
Method:	Least Squares		F-statistic:	891.1		
Date:	Fri, 06 Mar 2026		Prob (F-statistic):	2.87e-68		
Time:	06:05:05		Log-Likelihood:	111.24		
No. Observations:	112		AIC:	-216.5		
Df Residuals:	109		BIC:	-208.3		
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	1.1797	0.052	22.716	0.000	1.077	1.283
height	-0.0534	0.002	-32.956	0.000	-0.057	-0.050
width	0.0485	0.002	26.522	0.000	0.045	0.052
=====						
Omnibus:	37.934		Durbin-Watson:	1.378		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	67.608		
Skew:	1.493		Prob(JB):	2.08e-15		
Kurtosis:	5.360		Cond. No.	183.		
=====						

The R^2 value indicates that 94.24% of the variance in aspect ratio is explained by the height and width. While the adjusted R^2 differ from R^2 by only 0.0011 suggesting that the features are useful. The F-statistic is high suggest that the model fits the data well. Prob (F-statistic) is low suggesting the relationship between aspect ratio and the features are not due to random chance. In conclusion, all metrics suggest that this model is strong and fits the data well

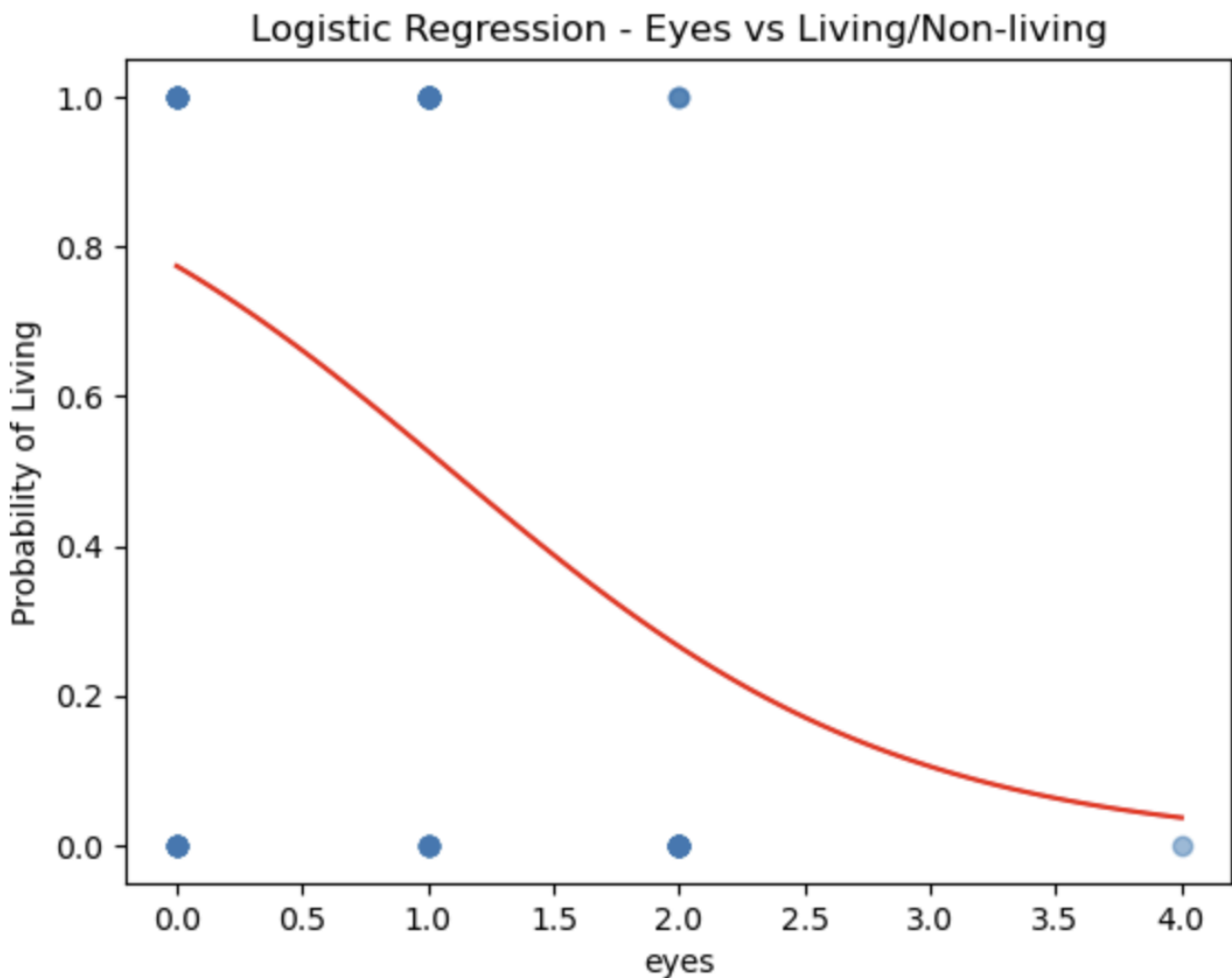
Section 4.3

Logistic regression was fitted using the **eyes** feature as it was the most significant discriminator $p = 0.000021$.

Logit Regression Results

Dep. Variable:	y	No. Observations:	112
Model:	Logit	Df Residuals:	110
Method:	MLE	Df Model:	1
Date:	Fri, 06 Mar 2026	Pseudo R-squ.:	0.1180
Time:	06:05:05	Log-Likelihood:	-68.472
converged:	True	LL-Null:	-77.632
Covariance Type:	nonrobust	LLR p-value:	1.867e-05

	coef	std err	z	P> z	[0.025	0.975]
const	1.2297	0.376	3.267	0.001	0.492	1.968
eyes	-1.1238	0.288	-3.898	0.000	-1.689	-0.559



The model showed that eyes has a negative coefficient (-1.1238), meaning more eyes, lower probability of being living. Both the intercept and eyes coefficient were statistically significant ($p < 0.05$). The overall model was significant, as LLR p-value confirming that eyes is a useful predictor for distinguishing living from non-living things.

Conclusions

Fulfilled the required answers and attached the required files

References